WHO Classification of Tumors of the Central Nervous System (2021)

Overview of the 5th Edition (WHO CNS5)

The 2021 Fifth Edition of the WHO Classification of Tumors of the Central Nervous System establishes molecular diagnosis as a core component of brain tumor classification alongside traditional histology. Tumors are graded from Grade 1 to Grade 4 based on histological features and molecular markers.

1. Gliomas, Glial Tumors, and Neuronal Tumors

Gliomas originate from glial support cells in the brain. They include Glioblastoma (IDH-wildtype, WHO Grade 4), Astrocytoma (IDH-mutant, WHO Grades 2, 3, or 4), and Oligodendroglioma (IDH-mutant and 1p/19q-codeleted, WHO Grades 2 or 3). Glioblastoma is the most aggressive primary brain tumor, characterized by microvascular proliferation and focal necrosis. IDH status (IDH1/IDH2 mutation) is a critical prognostic biomarker; IDH-mutant tumors generally have a significantly better prognosis than IDH-wildtype glioblastomas.

2. Meningiomas

Meningiomas arise from arachnoid cap cells of the meninges surrounding the brain and spinal cord. They are classified into WHO Grade 1 (benign, ~80-85% of cases), Grade 2 (atypical, ~15-20%), and Grade 3 (anaplastic/malignant, 1-3%). Subtypes include meningothelial, fibrous, transitional, psammomatous, clear cell, chordoid, rhabdoid, and papillary. Management typically involves surgical resection (Simpson grading) or stereotactic radiosurgery for residual/recurrent cases.

3. Pituitary Tumors (PitNETs)

Pituitary neuroendocrine tumors (PitNETs / Pituitary Adenomas) originate in the anterior pituitary gland in the sellar region. They are mostly slow-growing benign neoplasms. Functional adenomas hypersecrete hormones (e.g., prolactinomas causing galactorrhea/amenorrhea, somatotroph adenomas causing acromegaly, corticotroph adenomas causing Cushing disease). Non-functioning pituitary adenomas present due to mass effect on adjacent structures, notably compressive visual field deficits (bitemporal hemianopsia) due to optic chiasm displacement.

4. WHO Grading Definitions

Grade 1: Low proliferative potential, discrete boundaries, surgical cure alone is often achievable.
Grade 2: Generally infiltrative, low mitotic activity, higher risk of recurrence.
Grade 3: Histological evidence of malignancy, nuclear atypia, elevated mitotic figures.
Grade 4: Highly malignant, cytologically atypical, rapid pre- and post-operative progression, microvascular proliferation, and/or necrosis.